

Distribution Shift Results - SVGP

Skin Lesion 26 03 17 16 10 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J3 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.594 ± 0.003	0.584 ± 0.001	0.630 ± 0.001	0.616 ± 0.003	0.529 ± 0.142	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001
MAE	639.6 ± 20.7	354.9 ± 21.4	356.0 ± 21.8	350.4 ± 22.9	349.3 ± 22.8	431.5 ± 60.1	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0
RMSE	817.3 ± 35.0	538.1 ± 30.5	538.5 ± 31.5	540.0 ± 33.3	537.2 ± 33.7	600.9 ± 57.7	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5
NLL	1.501 ± 0.048	1.206 ± 0.121	1.206 ± 0.121	1.160 ± 0.117	1.161 ± 0.115	7.781 ± 6.624	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407
QCE 95	-0.029 ± 0.011	-0.102 ± 0.019	-0.100 ± 0.022	-0.084 ± 0.019	-0.089 ± 0.013	-0.225 ± 0.053	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019
PI 95 μ	3.920 ± 0.000	1.750 ± 0.020	1.748 ± 0.005	1.844 ± 0.009	1.819 ± 0.017	1.387 ± 0.059	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016
PI 95 σ	0.000 ± 0.000	0.131 ± 0.005	0.131 ± 0.006	0.112 ± 0.003	0.108 ± 0.001	0.832 ± 0.073	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004

J4 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.557 ± 0.003	0.543 ± 0.007	0.597 ± 0.001	0.578 ± 0.010	0.091 ± 0.002	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001
MAE	639.6 ± 20.7	335.0 ± 21.8	334.4 ± 20.6	339.1 ± 23.5	330.5 ± 23.8	296.6 ± 24.8	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0
RMSE	817.3 ± 35.0	530.2 ± 33.0	530.1 ± 33.1	540.0 ± 35.0	527.6 ± 35.4	479.5 ± 29.1	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5
NLL	1.501 ± 0.048	1.300 ± 0.125	1.283 ± 0.140	1.235 ± 0.117	1.203 ± 0.129	80.359 ± 31.322	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407
QCE 95	-0.029 ± 0.011	-0.104 ± 0.025	-0.104 ± 0.034	-0.096 ± 0.023	-0.091 ± 0.024	-0.723 ± 0.032	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019
PI 95 μ	3.920 ± 0.000	1.667 ± 0.011	1.674 ± 0.012	1.772 ± 0.007	1.758 ± 0.006	0.192 ± 0.012	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016
PI 95 σ	0.000 ± 0.000	0.126 ± 0.008	0.121 ± 0.009	0.113 ± 0.003	0.103 ± 0.003	0.114 ± 0.007	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004

J5 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.681 ± 0.002	0.688 ± 0.073	0.710 ± 0.001	0.684 ± 0.016	0.143 ± 0.002	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001
MAE	639.6 ± 20.7	388.0 ± 21.9	382.2 ± 24.2	390.0 ± 21.1	375.4 ± 23.8	355.6 ± 29.1	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0
RMSE	817.3 ± 35.0	588.3 ± 37.3	579.5 ± 35.6	593.0 ± 36.5	575.7 ± 39.3	555.1 ± 38.1	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5
NLL	1.501 ± 0.048	1.431 ± 0.140	1.396 ± 0.152	1.387 ± 0.129	1.350 ± 0.144	154.421 ± 60.393	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407
QCE 95	-0.029 ± 0.011	-0.124 ± 0.019	-0.114 ± 0.023	-0.119 ± 0.018	-0.106 ± 0.022	-0.784 ± 0.042	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019
PI 95 μ	3.920 ± 0.000	1.847 ± 0.009	1.835 ± 0.023	1.931 ± 0.002	1.894 ± 0.015	0.172 ± 0.038	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016
PI 95 σ	0.000 ± 0.000	0.115 ± 0.009	0.117 ± 0.007	0.116 ± 0.003	0.102 ± 0.006	0.082 ± 0.012	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004

Skin Lesion 26 03 17 16 10 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J3 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	0.681 ± 0.000	0.679 ± 0.000	0.726 ± 0.001	0.724 ± 0.001	0.694 ± 0.004	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001				
MAE	639.6 ± 20.7	377.9 ± 25.1	377.5 ± 24.9	376.5 ± 25.9	375.3 ± 25.1	500.0 ± 55.7	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0				
RMSE	817.3 ± 35.0	573.9 ± 33.6	573.3 ± 33.5	579.2 ± 35.6	577.6 ± 34.6	671.1 ± 73.4	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5				
NLL	1.501 ± 0.048	1.228 ± 0.117	1.227 ± 0.118	1.183 ± 0.105	1.181 ± 0.105	7.138 ± 6.246	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407				
QCE 95	-0.029 ± 0.011	-0.100 ± 0.021	-0.100 ± 0.021	-0.096 ± 0.022	-0.096 ± 0.023	-0.328 ± 0.173	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019				
PI 95 μ	3.920 ± 0.000	1.887 ± 0.005	1.884 ± 0.006	1.982 ± 0.006	1.978 ± 0.013	1.382 ± 0.220	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016				
PI 95 σ	0.000 ± 0.000	0.124 ± 0.006	0.124 ± 0.006	0.105 ± 0.004	0.104 ± 0.004	0.732 ± 0.137	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004				

J4 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	0.631 ± 0.001	0.624 ± 0.001	0.692 ± 0.003	0.684 ± 0.002	0.777 ± 0.116	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001				
MAE	639.6 ± 20.7	361.1 ± 25.5	359.3 ± 25.3	369.0 ± 26.3	366.6 ± 25.1	534.4 ± 68.2	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0				
RMSE	817.3 ± 35.0	568.2 ± 34.7	566.2 ± 34.7	581.4 ± 34.5	579.1 ± 34.1	701.3 ± 90.3	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5				
NLL	1.501 ± 0.048	1.308 ± 0.120	1.304 ± 0.119	1.241 ± 0.106	1.241 ± 0.105	22.307 ± 26.697	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407				
QCE 95	-0.029 ± 0.011	-0.105 ± 0.027	-0.104 ± 0.025	-0.095 ± 0.019	-0.096 ± 0.023	-0.448 ± 0.221	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019				
PI 95 μ	3.920 ± 0.000	1.781 ± 0.010	1.776 ± 0.010	1.919 ± 0.020	1.906 ± 0.013	1.230 ± 0.221	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016				
PI 95 σ	0.000 ± 0.000	0.128 ± 0.007	0.128 ± 0.007	0.112 ± 0.004	0.111 ± 0.003	0.742 ± 0.288	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004				

J5 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	0.710 ± 0.001	0.695 ± 0.001	0.763 ± 0.001	0.747 ± 0.001	0.307 ± 0.126	0.741 ± 0.020	0.715 ± 0.018	0.748 ± 0.006	0.718 ± 0.004	0.167 ± 0.001	1.276 ± 0.726	0.911 ± 0.039	1.002 ± 0.078	0.916 ± 0.035	0.129 ± 0.001				
MAE	639.6 ± 20.7	402.7 ± 20.5	396.4 ± 18.5	414.5 ± 20.1	408.5 ± 21.2	373.2 ± 46.6	340.0 ± 18.5	335.9 ± 19.6	341.9 ± 23.3	334.8 ± 21.7	312.0 ± 18.8	399.1 ± 43.9	381.8 ± 18.2	400.2 ± 28.1	378.1 ± 9.5	334.9 ± 12.0				
RMSE	817.3 ± 35.0	615.0 ± 35.9	606.9 ± 33.1	630.4 ± 35.3	623.1 ± 36.4	554.7 ± 49.5	515.4 ± 38.9	513.2 ± 36.3	529.2 ± 45.7	516.9 ± 40.3	485.8 ± 34.5	584.8 ± 73.6	555.8 ± 34.0	578.1 ± 33.5	555.7 ± 20.5	510.2 ± 21.5				
NLL	1.501 ± 0.048	1.450 ± 0.115	1.436 ± 0.117	1.384 ± 0.098	1.374 ± 0.097	6.613 ± 3.193	1.126 ± 0.128	1.128 ± 0.139	1.149 ± 0.145	1.133 ± 0.144	75.017 ± 32.110	1.178 ± 0.124	1.153 ± 0.093	1.180 ± 0.087	1.148 ± 0.072	148.111 ± 82.407				
QCE 95	-0.029 ± 0.011	-0.119 ± 0.014	-0.120 ± 0.018	-0.112 ± 0.017	-0.112 ± 0.017	-0.308 ± 1.307	-0.076 ± 0.022	-0.083 ± 0.026	-0.080 ± 0.023	-0.080 ± 0.022	-0.696 ± 0.047	-0.060 ± 0.010	-0.060 ± 0.015	-0.054 ± 0.008	-0.060 ± 0.018	-0.730 ± 0.019				
PI 95 μ	3.920 ± 0.000	1.905 ± 0.004	1.884 ± 0.006	2.040 ± 0.007	2.020 ± 0.007	1.101 ± 0.349	1.926 ± 0.007	1.908 ± 0.019	1.960 ± 0.008	1.926 ± 0.004	0.208 ± 0.032	2.464 ± 0.231	2.319 ± 0.043	2.477 ± 0.107	2.321 ± 0.051	0.187 ± 0.016				
PI 95 σ	0.000 ± 0.000	0.130 ± 0.009	0.128 ± 0.008	0.125 ± 0.004	0.123 ± 0.005	0.691 ± 0.242	0.086 ± 0.010	0.081 ± 0.010	0.088 ± 0.007	0.083 ± 0.007	0.060 ± 0.005	0.083 ± 0.012	0.069 ± 0.006	0.074 ± 0.009	0.068 ± 0.008	0.083 ± 0.004				

Skin Lesion 26 03 17 16 10 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J3 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.594 ± 0.003	0.584 ± 0.001	0.630 ± 0.001	0.616 ± 0.003	0.529 ± 0.142	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	354.9 ± 21.4	356.0 ± 21.8	350.4 ± 22.9	349.3 ± 22.8	431.5 ± 60.1	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.0 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	538.1 ± 30.5	538.5 ± 31.5	540.0 ± 33.3	537.2 ± 33.7	600.9 ± 57.7	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.206 ± 0.121	1.206 ± 0.121	1.160 ± 0.117	1.161 ± 0.115	7.781 ± 0.624	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.440 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.102 ± 0.019	-0.100 ± 0.022	-0.084 ± 0.019	-0.089 ± 0.013	-0.225 ± 0.053	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.750 ± 0.020	1.748 ± 0.005	1.844 ± 0.009	1.819 ± 0.017	1.387 ± 0.059	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.131 ± 0.005	0.131 ± 0.006	0.112 ± 0.003	0.108 ± 0.001	0.832 ± 0.073	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

J4 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.557 ± 0.003	0.543 ± 0.007	0.597 ± 0.001	0.578 ± 0.010	0.091 ± 0.002	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	335.0 ± 21.8	334.4 ± 20.6	330.1 ± 23.5	330.5 ± 23.8	296.6 ± 24.8	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.0 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	530.2 ± 33.0	530.1 ± 33.1	540.0 ± 35.0	527.6 ± 35.4	479.5 ± 29.1	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.300 ± 0.125	1.283 ± 0.140	1.235 ± 0.117	1.203 ± 0.120	80.359 ± 31.322	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.440 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.104 ± 0.025	-0.104 ± 0.034	-0.096 ± 0.023	-0.091 ± 0.024	-0.723 ± 0.032	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.667 ± 0.011	1.674 ± 0.012	1.772 ± 0.007	1.758 ± 0.006	0.192 ± 0.022	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.126 ± 0.008	0.121 ± 0.009	0.113 ± 0.003	0.103 ± 0.003	0.114 ± 0.007	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.681 ± 0.002	0.688 ± 0.073	0.710 ± 0.001	0.684 ± 0.016	0.143 ± 0.002	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	388.0 ± 21.9	382.2 ± 24.2	390.0 ± 21.1	375.4 ± 23.8	355.6 ± 29.1	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.0 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	588.3 ± 37.3	579.5 ± 36.5	593.0 ± 36.5	575.7 ± 39.3	555.1 ± 38.1	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.431 ± 0.140	1.396 ± 0.152	1.387 ± 0.129	1.350 ± 0.144	154.421 ± 60.393	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.440 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.124 ± 0.019	-0.114 ± 0.022	-0.119 ± 0.018	-0.106 ± 0.022	-0.784 ± 0.042	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.847 ± 0.009	1.835 ± 0.023	1.931 ± 0.002	1.894 ± 0.015	0.172 ± 0.038	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.115 ± 0.009	0.117 ± 0.007	0.116 ± 0.003	0.102 ± 0.006	0.082 ± 0.012	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

Skin Lesion 26 03 17 16 10 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J3 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.681 ± 0.000	0.679 ± 0.000	0.726 ± 0.001	0.724 ± 0.001	0.694 ± 0.004	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	377.9 ± 25.1	377.5 ± 24.9	376.5 ± 25.9	375.3 ± 25.1	500.0 ± 55.7	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.9 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	573.9 ± 33.6	573.3 ± 33.5	579.2 ± 35.6	577.6 ± 34.6	671.1 ± 73.4	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.228 ± 0.117	1.227 ± 0.118	1.183 ± 0.105	1.181 ± 0.105	7.138 ± 0.246	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.449 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.100 ± 0.021	-0.100 ± 0.021	-0.096 ± 0.022	-0.096 ± 0.023	-0.328 ± 0.173	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.887 ± 0.005	1.884 ± 0.006	1.982 ± 0.006	1.978 ± 0.013	1.382 ± 0.220	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.124 ± 0.006	0.124 ± 0.006	0.105 ± 0.004	0.104 ± 0.004	0.732 ± 0.137	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

J4 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.631 ± 0.001	0.624 ± 0.001	0.692 ± 0.003	0.684 ± 0.002	0.777 ± 0.116	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	361.1 ± 25.5	359.3 ± 25.3	369.0 ± 26.3	366.6 ± 25.1	534.4 ± 68.2	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.9 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	568.2 ± 34.7	566.2 ± 34.7	581.4 ± 34.5	579.1 ± 34.1	701.3 ± 90.3	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.308 ± 0.120	1.304 ± 0.119	1.241 ± 0.106	1.241 ± 0.105	22.307 ± 26.697	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.449 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.105 ± 0.027	-0.104 ± 0.025	-0.095 ± 0.019	-0.096 ± 0.023	-0.448 ± 0.221	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.781 ± 0.010	1.776 ± 0.010	1.919 ± 0.020	1.906 ± 0.013	1.230 ± 0.290	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.128 ± 0.007	0.128 ± 0.007	0.112 ± 0.004	0.111 ± 0.003	0.742 ± 0.288	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.710 ± 0.001	0.695 ± 0.001	0.763 ± 0.001	0.747 ± 0.001	0.307 ± 0.126	1.135 ± 0.011	1.128 ± 0.060	1.144 ± 0.007	1.144 ± 0.044	0.515 ± 0.009	1.016 ± 0.006	1.054 ± 0.120	1.036 ± 0.013	1.071 ± 0.112	0.321 ± 0.004
MAE	639.6 ± 20.7	402.7 ± 20.5	396.4 ± 18.5	414.5 ± 20.1	408.5 ± 21.2	373.2 ± 46.6	541.2 ± 22.8	533.5 ± 38.4	540.4 ± 33.3	542.1 ± 37.2	542.8 ± 28.5	481.5 ± 23.6	480.9 ± 32.0	483.7 ± 22.9	490.5 ± 41.0	475.7 ± 25.1
RMSE	817.3 ± 35.0	615.0 ± 35.9	606.9 ± 33.1	630.4 ± 35.3	623.1 ± 36.4	554.7 ± 49.5	760.4 ± 45.1	747.6 ± 56.0	754.7 ± 57.1	756.6 ± 62.6	755.7 ± 47.6	691.4 ± 39.0	687.4 ± 40.3	694.2 ± 37.6	696.9 ± 65.0	685.9 ± 43.0
NLL	1.501 ± 0.048	1.450 ± 0.115	1.436 ± 0.117	1.384 ± 0.098	1.374 ± 0.097	6.613 ± 3.193	1.540 ± 0.110	1.464 ± 0.119	1.528 ± 0.126	1.481 ± 0.128	41.671 ± 25.282	1.449 ± 0.106	1.396 ± 0.098	1.443 ± 0.107	1.424 ± 0.141	113.553 ± 37.665
QCE 95	-0.029 ± 0.011	-0.119 ± 0.014	-0.120 ± 0.018	-0.112 ± 0.017	-0.112 ± 0.017	-0.308 ± 0.107	-0.104 ± 0.027	-0.100 ± 0.017	-0.098 ± 0.023	-0.096 ± 0.027	-0.684 ± 0.085	-0.098 ± 0.017	-0.096 ± 0.019	-0.092 ± 0.021	-0.092 ± 0.029	-0.775 ± 0.036
PI 95 μ	3.920 ± 0.000	1.905 ± 0.004	1.884 ± 0.006	2.040 ± 0.007	2.020 ± 0.007	1.101 ± 0.349	2.719 ± 0.004	2.713 ± 0.082	2.738 ± 0.009	2.819 ± 0.033	0.486 ± 0.118	2.470 ± 0.013	2.500 ± 0.086	2.519 ± 0.016	2.538 ± 0.056	0.258 ± 0.063
PI 95 σ	0.000 ± 0.000	0.130 ± 0.009	0.128 ± 0.008	0.125 ± 0.004	0.123 ± 0.005	0.691 ± 0.242	0.138 ± 0.028	0.129 ± 0.014	0.143 ± 0.024	0.113 ± 0.013	0.084 ± 0.009	0.218 ± 0.055	0.172 ± 0.039	0.207 ± 0.043	0.163 ± 0.032	0.094 ± 0.014

Histology Nuclei 26 03 17 15 09 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J3 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.051 ± 0.001	1.048 ± 0.000	1.040 ± 0.001	1.039 ± 0.001	0.800 ± 0.045	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	297.2 ± 9.0	297.2 ± 9.0	298.6 ± 10.3	298.7 ± 10.2	446.9 ± 39.9	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	397.6 ± 9.3	396.9 ± 9.3	396.7 ± 10.2	396.3 ± 9.9	523.3 ± 38.6	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.844 ± 0.021	0.843 ± 0.022	0.839 ± 0.022	0.838 ± 0.023	7.298 ± 6.876	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.008 ± 0.006	0.009 ± 0.007	0.012 ± 0.010	0.012 ± 0.010	-0.388 ± 0.174	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.566 ± 0.004	2.559 ± 0.004	2.557 ± 0.005	2.557 ± 0.004	1.291 ± 0.297	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.017 ± 0.001	0.017 ± 0.001	0.021 ± 0.001	0.021 ± 0.002	0.401 ± 0.117	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

J4 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.013 ± 0.000	1.003 ± 0.001	1.007 ± 0.001	0.999 ± 0.001	0.365 ± 0.010	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	291.9 ± 7.3	290.7 ± 11.0	294.8 ± 9.1	292.7 ± 11.6	332.6 ± 6.6	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	389.5 ± 8.7	387.3 ± 10.9	389.3 ± 9.5	390.9 ± 12.2	475.7 ± 5.7	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.826 ± 0.021	0.821 ± 0.023	0.827 ± 0.023	0.825 ± 0.025	49.284 ± 16.053	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.069 ± 0.009	0.017 ± 0.009	0.005 ± 0.013	0.011 ± 0.012	-0.587 ± 0.059	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.485 ± 0.001	2.481 ± 0.002	2.492 ± 0.005	2.487 ± 0.004	0.378 ± 0.087	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.018 ± 0.002	0.022 ± 0.001	0.024 ± 0.002	0.025 ± 0.002	0.153 ± 0.058	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

J5 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.996 ± 0.001	0.968 ± 0.002	0.990 ± 0.001	0.968 ± 0.001	0.362 ± 0.003	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	301.0 ± 10.9	305.4 ± 13.2	302.0 ± 10.3	303.7 ± 11.3	312.9 ± 8.3	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	391.9 ± 13.7	394.0 ± 15.4	399.0 ± 13.2	395.6 ± 14.8	437.1 ± 9.2	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.833 ± 0.027	0.834 ± 0.029	0.835 ± 0.026	0.836 ± 0.026	71.950 ± 45.908	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.009 ± 0.015	0.012 ± 0.008	0.001 ± 0.012	0.012 ± 0.010	-0.706 ± 0.050	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.449 ± 0.003	2.420 ± 0.007	2.462 ± 0.002	2.439 ± 0.005	0.247 ± 0.056	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.018 ± 0.001	0.032 ± 0.004	0.025 ± 0.001	0.031 ± 0.002	0.062 ± 0.010	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

Histology Nuclei 26 03 17 15 09 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J3 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.073 ± 0.000	1.073 ± 0.000	1.066 ± 0.000	1.066 ± 0.000	0.936 ± 0.027	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	295.7 ± 11.3	295.6 ± 11.3	291.3 ± 13.2	291.3 ± 13.0	509.6 ± 18.7	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	394.2 ± 11.4	394.2 ± 11.2	381.3 ± 12.5	381.2 ± 12.4	589.4 ± 17.0	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.846 ± 0.024	0.846 ± 0.023	0.834 ± 0.024	0.833 ± 0.024	58.080 ± 75.395	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.015 ± 0.007	0.015 ± 0.007	0.023 ± 0.009	0.023 ± 0.009	-0.684 ± 0.070	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.608 ± 0.003	2.608 ± 0.003	2.614 ± 0.002	2.613 ± 0.002	0.810 ± 0.200	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.018 ± 0.001	0.018 ± 0.001	0.242 ± 0.111	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

J4 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.044 ± 0.001	1.043 ± 0.001	1.040 ± 0.000	1.040 ± 0.000	0.785 ± 0.070	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	290.5 ± 9.9	290.5 ± 9.9	287.7 ± 12.7	287.8 ± 12.7	447.3 ± 17.9	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	379.4 ± 11.2	379.4 ± 11.1	375.1 ± 12.3	375.1 ± 12.4	524.8 ± 23.2	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.824 ± 0.023	0.824 ± 0.022	0.819 ± 0.024	0.819 ± 0.024	5.936 ± 4.214	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.023 ± 0.010	0.023 ± 0.010	0.023 ± 0.010	0.023 ± 0.010	-0.411 ± 0.134	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.538 ± 0.005	2.538 ± 0.005	2.557 ± 0.002	2.557 ± 0.002	1.269 ± 0.209	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.018 ± 0.002	0.018 ± 0.002	0.414 ± 0.166	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

J5 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.026 ± 0.000	1.023 ± 0.000	1.016 ± 0.000	1.013 ± 0.001	0.527 ± 0.030	1.021 ± 0.112	0.970 ± 0.029	0.971 ± 0.006	0.945 ± 0.001	0.342 ± 0.001	1.040 ± 0.005	1.070 ± 0.082	1.037 ± 0.006	1.058 ± 0.033	0.314 ± 0.002
MAE	540.8 ± 18.0	299.6 ± 11.2	299.5 ± 11.0	294.9 ± 11.1	294.7 ± 10.8	347.8 ± 17.3	390.1 ± 99.4	328.7 ± 12.6	360.9 ± 22.7	340.2 ± 1.3	358.9 ± 12.2	348.7 ± 22.2	341.2 ± 10.9	342.3 ± 10.5	335.7 ± 21.9	337.2 ± 18.4
RMSE	623.4 ± 14.1	381.6 ± 15.0	381.5 ± 15.0	379.5 ± 14.0	379.1 ± 13.9	428.4 ± 18.7	509.1 ± 97.3	450.6 ± 24.8	492.6 ± 22.8	470.3 ± 10.2	479.1 ± 13.7	496.5 ± 20.5	488.0 ± 13.0	489.5 ± 6.9	480.5 ± 25.5	466.9 ± 11.7
NLL	1.308 ± 0.018	0.828 ± 0.027	0.827 ± 0.027	0.823 ± 0.026	0.823 ± 0.025	5.146 ± 2.183	0.872 ± 0.025	0.867 ± 0.025	0.884 ± 0.034	0.881 ± 0.025	63.454 ± 21.263	0.940 ± 0.019	0.929 ± 0.032	0.933 ± 0.019	0.920 ± 0.028	350.853 ± 167.853
QCE 95	0.047 ± 0.003	0.019 ± 0.007	0.020 ± 0.007	0.020 ± 0.009	0.020 ± 0.009	-0.294 ± 0.057	-0.055 ± 0.045	-0.028 ± 0.013	-0.047 ± 0.018	-0.037 ± 0.008	-0.756 ± 0.042	-0.035 ± 0.018	-0.027 ± 0.010	-0.029 ± 0.009	-0.028 ± 0.012	-0.829 ± 0.046
PI 95 μ	3.920 ± 0.000	2.500 ± 0.002	2.496 ± 0.001	2.508 ± 0.002	2.504 ± 0.004	1.171 ± 0.126	2.481 ± 0.053	2.448 ± 0.028	2.478 ± 0.008	2.457 ± 0.015	0.266 ± 0.042	2.572 ± 0.009	2.621 ± 0.088	2.569 ± 0.009	2.580 ± 0.026	0.128 ± 0.033
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.020 ± 0.001	0.020 ± 0.001	0.472 ± 0.076	0.050 ± 0.003	0.052 ± 0.007	0.057 ± 0.005	0.060 ± 0.007	0.073 ± 0.008	0.065 ± 0.004	0.054 ± 0.004	0.067 ± 0.005	0.057 ± 0.002	0.051 ± 0.007

Histology Nuclei 26 03 17 15 09 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J3 - L8

Metric	Base					Scattering					DINOv2 Small					DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.051 ± 0.001	1.048 ± 0.000	1.040 ± 0.001	1.039 ± 0.001	0.800 ± 0.045	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.073	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	297.2 ± 9.0	297.2 ± 9.0	296.6 ± 10.3	298.7 ± 10.2	446.9 ± 39.9	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	397.6 ± 9.3	396.9 ± 9.3	396.7 ± 10.2	396.3 ± 9.9	523.3 ± 38.6	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.844 ± 0.021	0.843 ± 0.022	0.839 ± 0.022	0.838 ± 0.023	7.298 ± 6.876	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.008 ± 0.006	0.009 ± 0.007	0.012 ± 0.010	0.012 ± 0.010	-0.388 ± 0.174	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.013	-0.742 ± 0.010	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.013	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.566 ± 0.004	2.559 ± 0.004	2.557 ± 0.005	2.557 ± 0.004	1.291 ± 0.297	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.017 ± 0.001	0.017 ± 0.001	0.021 ± 0.001	0.021 ± 0.002	0.401 ± 0.117	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016

J4 - L8

Metric	Base					Scattering					DINOv2 Small					DINOv2 Base					
	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	
LOSS	0.000 ± 0.000	1.013 ± 0.000	1.003 ± 0.001	1.007 ± 0.001	0.999 ± 0.001	0.365 ± 0.010	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.093	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	291.9 ± 7.3	290.7 ± 11.0	294.8 ± 9.1	292.7 ± 11.6	332.6 ± 6.6	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	389.5 ± 8.7	382.3 ± 10.9	396.3 ± 9.5	390.9 ± 12.2	475.7 ± 5.7	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.826 ± 0.021	0.821 ± 0.023	0.827 ± 0.023	0.825 ± 0.025	49.284 ± 16.053	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.009 ± 0.009	0.017 ± 0.009	0.004 ± 0.013	0.011 ± 0.012	-0.587 ± 0.059	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.485 ± 0.001	2.481 ± 0.002	2.492 ± 0.005	2.487 ± 0.004	0.378 ± 0.087	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.018 ± 0.002	0.022 ± 0.001	0.024 ± 0.002	0.025 ± 0.002	0.153 ± 0.058	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016

J5 - L8

Metric	Base					Scattering					DINOv2 Small					DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.996 ± 0.001	0.968 ± 0.002	0.990 ± 0.001	0.968 ± 0.001	0.362 ± 0.003	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.093	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	301.0 ± 10.9	305.4 ± 13.2	302.0 ± 10.3	303.7 ± 11.3	312.9 ± 8.3	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	391.9 ± 13.7	394.0 ± 15.4	399.0 ± 13.2	395.6 ± 14.8	437.1 ± 9.2	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.833 ± 0.027	0.834 ± 0.029	0.835 ± 0.026	0.836 ± 0.026	71.980 ± 45.908	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.009 ± 0.015	0.012 ± 0.008	0.001 ± 0.012	0.012 ± 0.010	-0.706 ± 0.050	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.449 ± 0.003	2.420 ± 0.007	2.462 ± 0.002	2.439 ± 0.005	0.247 ± 0.056	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.018 ± 0.001	0.032 ± 0.004	0.025 ± 0.001	0.031 ± 0.002	0.062 ± 0.010	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.062 ± 0.016

Histology Nuclei 26 03 17 15 09 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J3 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.073 ± 0.000	1.073 ± 0.000	1.066 ± 0.000	1.066 ± 0.000	0.936 ± 0.027	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.093	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	295.7 ± 11.3	295.6 ± 11.3	291.3 ± 13.2	291.3 ± 13.0	509.6 ± 18.7	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	394.2 ± 11.4	394.2 ± 11.2	381.3 ± 12.5	381.2 ± 12.4	589.4 ± 17.0	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.846 ± 0.024	0.846 ± 0.023	0.834 ± 0.024	0.833 ± 0.024	58.080 ± 75.395	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.015 ± 0.007	0.015 ± 0.007	0.023 ± 0.009	0.023 ± 0.009	-0.684 ± 0.070	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.608 ± 0.003	2.608 ± 0.003	2.614 ± 0.002	2.613 ± 0.002	0.810 ± 0.200	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.018 ± 0.001	0.018 ± 0.001	0.242 ± 0.111	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.082 ± 0.016

J4 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.044 ± 0.001	1.043 ± 0.001	1.040 ± 0.000	1.040 ± 0.000	0.785 ± 0.070	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.093	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	290.5 ± 9.9	290.5 ± 9.9	287.7 ± 12.7	287.8 ± 12.7	447.3 ± 17.9	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	379.4 ± 11.2	379.4 ± 11.1	375.1 ± 12.3	375.1 ± 12.4	524.8 ± 23.2	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.824 ± 0.023	0.824 ± 0.022	0.819 ± 0.024	0.819 ± 0.024	5.936 ± 4.214	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.023 ± 0.010	0.023 ± 0.010	0.023 ± 0.010	0.023 ± 0.010	-0.411 ± 0.134	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.538 ± 0.005	2.538 ± 0.005	2.557 ± 0.002	2.557 ± 0.003	1.269 ± 0.209	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.018 ± 0.002	0.018 ± 0.002	0.414 ± 0.166	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.082 ± 0.016

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.026 ± 0.000	1.023 ± 0.000	1.016 ± 0.000	1.013 ± 0.001	0.527 ± 0.030	1.019 ± 0.004	1.048 ± 0.043	1.062 ± 0.093	1.036 ± 0.073	0.435 ± 0.005	0.961 ± 0.012	0.930 ± 0.033	0.952 ± 0.023	0.897 ± 0.007	0.331 ± 0.003
MAE	540.8 ± 18.0	299.6 ± 11.2	299.5 ± 11.0	294.9 ± 11.1	294.7 ± 10.8	347.8 ± 17.3	305.5 ± 13.4	310.0 ± 20.1	308.3 ± 16.5	307.7 ± 12.5	336.3 ± 17.2	291.2 ± 23.1	287.6 ± 12.4	296.6 ± 10.6	285.5 ± 13.4	293.8 ± 9.1
RMSE	623.4 ± 14.1	381.6 ± 15.0	381.5 ± 15.0	379.5 ± 14.0	379.1 ± 13.9	428.4 ± 18.7	416.0 ± 19.0	419.4 ± 21.7	421.8 ± 25.8	421.1 ± 19.2	444.4 ± 17.4	391.0 ± 25.5	389.2 ± 24.7	404.6 ± 16.9	392.0 ± 19.5	393.7 ± 13.8
NLL	1.308 ± 0.018	0.828 ± 0.027	0.827 ± 0.027	0.823 ± 0.026	0.822 ± 0.025	5.146 ± 2.183	0.856 ± 0.038	0.859 ± 0.039	0.848 ± 0.039	0.843 ± 0.034	87.855 ± 28.530	0.803 ± 0.041	0.778 ± 0.041	0.801 ± 0.031	0.774 ± 0.043	72.259 ± 18.251
QCE 95	0.047 ± 0.003	0.019 ± 0.007	0.020 ± 0.007	0.020 ± 0.009	0.020 ± 0.009	-0.294 ± 0.057	0.004 ± 0.009	0.004 ± 0.007	-0.002 ± 0.013	0.002 ± 0.007	-0.795 ± 0.036	0.004 ± 0.011	0.004 ± 0.010	-0.002 ± 0.013	0.002 ± 0.014	-0.742 ± 0.010
PI 95 μ	3.920 ± 0.000	2.500 ± 0.002	2.496 ± 0.001	2.508 ± 0.002	2.504 ± 0.004	1.171 ± 0.126	2.536 ± 0.009	2.588 ± 0.046	2.504 ± 0.027	2.545 ± 0.031	0.204 ± 0.033	2.437 ± 0.012	2.378 ± 0.025	2.426 ± 0.043	2.357 ± 0.026	0.221 ± 0.020
PI 95 σ	0.000 ± 0.000	0.014 ± 0.001	0.014 ± 0.001	0.020 ± 0.001	0.020 ± 0.001	0.472 ± 0.076	0.072 ± 0.014	0.069 ± 0.029	0.082 ± 0.004	0.105 ± 0.033	0.060 ± 0.016	0.113 ± 0.013	0.125 ± 0.034	0.120 ± 0.007	0.121 ± 0.013	0.082 ± 0.016

Poverty 26 03 18 07 02 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J6 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.487 ± 0.011	1.487 ± 0.006	1.468 ± 0.114	1.127 ± 0.045	0.290 ± 0.003	0.920 ± 0.002	0.874 ± 0.001	0.936 ± 0.001	0.885 ± 0.002	0.208 ± 0.004	0.944 ± 0.001	0.957 ± 0.054	0.961 ± 0.004	0.942 ± 0.035	0.114 ± 0.008
MAE	0.798 ± 0.025	0.798 ± 0.025	0.798 ± 0.025	0.722 ± 0.084	0.460 ± 0.026	0.423 ± 0.008	0.422 ± 0.007	0.398 ± 0.007	0.425 ± 0.007	0.398 ± 0.007	0.389 ± 0.010	0.413 ± 0.009	0.408 ± 0.015	0.418 ± 0.007	0.386 ± 0.027	0.411 ± 0.018
RMSE	0.903 ± 0.020	0.903 ± 0.020	0.903 ± 0.020	0.821 ± 0.087	0.568 ± 0.033	0.545 ± 0.011	0.528 ± 0.006	0.498 ± 0.007	0.532 ± 0.006	0.498 ± 0.006	0.498 ± 0.006	0.524 ± 0.015	0.517 ± 0.015	0.529 ± 0.016	0.496 ± 0.027	0.524 ± 0.016
NLL	1.541 ± 0.027	1.378 ± 0.022	1.373 ± 0.023	1.262 ± 0.081	0.985 ± 0.052	80.981 ± 18.642	0.910 ± 0.014	0.864 ± 0.023	0.916 ± 0.013	0.866 ± 0.020	75.968 ± 10.258	0.935 ± 0.035	0.899 ± 0.035	0.940 ± 0.035	0.884 ± 0.046	36.221 ± 12.337
QCE 95	0.014 ± 0.009	-0.004 ± 0.009	-0.004 ± 0.009	0.007 ± 0.020	0.017 ± 0.009	-0.798 ± 0.021	-0.015 ± 0.007	-0.007 ± 0.010	-0.012 ± 0.006	-0.002 ± 0.007	-0.791 ± 0.019	-0.016 ± 0.017	-0.012 ± 0.012	-0.014 ± 0.017	-0.008 ± 0.014	-0.749 ± 0.054
PI 95 μ	3.920 ± 0.000	3.563 ± 0.013	3.561 ± 0.008	3.398 ± 0.104	2.814 ± 0.164	0.244 ± 0.016	2.381 ± 0.008	2.297 ± 0.006	2.425 ± 0.004	2.324 ± 0.011	0.235 ± 0.022	2.432 ± 0.007	2.412 ± 0.026	2.475 ± 0.019	2.415 ± 0.020	0.337 ± 0.044
PI 95 σ	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.033 ± 0.031	0.121 ± 0.019	0.140 ± 0.009	0.088 ± 0.002	0.076 ± 0.002	0.079 ± 0.002	0.066 ± 0.001	0.098 ± 0.011	0.058 ± 0.002	0.039 ± 0.002	0.053 ± 0.002	0.033 ± 0.003	0.119 ± 0.013

J7 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.475 ± 0.000	1.475 ± 0.000	1.446 ± 0.067	1.476 ± 0.001	0.328 ± 0.001	0.920 ± 0.002	0.874 ± 0.001	0.936 ± 0.001	0.885 ± 0.002	0.208 ± 0.004	0.944 ± 0.001	0.957 ± 0.054	0.961 ± 0.004	0.942 ± 0.035	0.114 ± 0.008
MAE	0.798 ± 0.025	0.798 ± 0.025	0.798 ± 0.025	0.756 ± 0.078	0.798 ± 0.025	0.441 ± 0.013	0.422 ± 0.007	0.398 ± 0.007	0.425 ± 0.007	0.398 ± 0.007	0.389 ± 0.010	0.413 ± 0.009	0.408 ± 0.015	0.418 ± 0.007	0.386 ± 0.027	0.411 ± 0.018
RMSE	0.903 ± 0.020	0.903 ± 0.020	0.903 ± 0.020	0.861 ± 0.081	0.903 ± 0.020	0.563 ± 0.016	0.528 ± 0.006	0.498 ± 0.007	0.532 ± 0.006	0.498 ± 0.006	0.498 ± 0.006	0.524 ± 0.015	0.517 ± 0.015	0.529 ± 0.016	0.496 ± 0.027	0.524 ± 0.016
NLL	1.541 ± 0.027	1.380 ± 0.028	1.384 ± 0.033	1.318 ± 0.069	1.355 ± 0.020	282.055 ± 44.924	0.910 ± 0.014	0.864 ± 0.023	0.916 ± 0.013	0.866 ± 0.020	75.968 ± 10.258	0.935 ± 0.035	0.899 ± 0.035	0.940 ± 0.035	0.884 ± 0.046	36.221 ± 12.337
QCE 95	0.014 ± 0.009	-0.004 ± 0.009	-0.004 ± 0.009	0.002 ± 0.012	-0.004 ± 0.009	-0.866 ± 0.015	-0.015 ± 0.007	-0.007 ± 0.010	-0.012 ± 0.006	-0.002 ± 0.007	-0.791 ± 0.019	-0.016 ± 0.017	-0.012 ± 0.012	-0.014 ± 0.017	-0.008 ± 0.014	-0.749 ± 0.054
PI 95 μ	3.920 ± 0.000	3.577 ± 0.000	3.577 ± 0.000	3.514 ± 0.138	3.576 ± 0.001	0.152 ± 0.009	2.381 ± 0.008	2.297 ± 0.006	2.425 ± 0.004	2.324 ± 0.011	0.235 ± 0.022	2.432 ± 0.007	2.412 ± 0.026	2.475 ± 0.019	2.415 ± 0.020	0.337 ± 0.044
PI 95 σ	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.006 ± 0.012	0.000 ± 0.000	0.106 ± 0.010	0.088 ± 0.002	0.076 ± 0.002	0.079 ± 0.002	0.066 ± 0.001	0.098 ± 0.011	0.058 ± 0.002	0.039 ± 0.002	0.053 ± 0.002	0.033 ± 0.003	0.119 ± 0.013

Poverty 26 03 18 07 02 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J6 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.330 ± 0.109	1.326 ± 0.207	1.314 ± 0.171	1.113 ± 0.031	0.286 ± 0.008	0.920 ± 0.002	0.874 ± 0.001	0.936 ± 0.001	0.885 ± 0.002	0.208 ± 0.004	0.944 ± 0.001	0.957 ± 0.054	0.961 ± 0.004	0.942 ± 0.035	0.114 ± 0.008
MAE	0.798 ± 0.025	0.630 ± 0.054	0.678 ± 0.244	0.650 ± 0.095	0.472 ± 0.011	0.427 ± 0.006	0.422 ± 0.007	0.398 ± 0.007	0.425 ± 0.007	0.398 ± 0.007	0.389 ± 0.010	0.413 ± 0.009	0.408 ± 0.015	0.418 ± 0.007	0.386 ± 0.027	0.411 ± 0.018
RMSE	0.903 ± 0.020	0.745 ± 0.046	0.798 ± 0.267	0.753 ± 0.091	0.576 ± 0.015	0.533 ± 0.009	0.528 ± 0.006	0.498 ± 0.007	0.532 ± 0.006	0.498 ± 0.006	0.498 ± 0.006	0.524 ± 0.015	0.517 ± 0.015	0.529 ± 0.016	0.496 ± 0.027	0.524 ± 0.016
NLL	1.541 ± 0.027	1.216 ± 0.070	1.037 ± 0.064	1.188 ± 0.061	0.986 ± 0.025	231.332 ± 190.047	0.910 ± 0.014	0.864 ± 0.023	0.916 ± 0.013	0.866 ± 0.020	75.968 ± 10.258	0.935 ± 0.035	0.899 ± 0.035	0.940 ± 0.035	0.884 ± 0.046	36.221 ± 12.337
QCE 95	0.014 ± 0.009	-0.017 ± 0.033	-0.071 ± 0.170	0.004 ± 0.007	0.016 ± 0.010	-0.820 ± 0.052	-0.015 ± 0.007	-0.007 ± 0.010	-0.012 ± 0.006	-0.002 ± 0.007	-0.791 ± 0.019	-0.016 ± 0.017	-0.012 ± 0.012	-0.014 ± 0.017	-0.008 ± 0.014	-0.749 ± 0.054
PI 95 μ	3.920 ± 0.000	3.200 ± 0.268	3.087 ± 0.175	3.223 ± 0.219	2.815 ± 0.072	0.229 ± 0.098	2.381 ± 0.008	2.297 ± 0.006	2.425 ± 0.004	2.324 ± 0.011	0.235 ± 0.022	2.432 ± 0.007	2.412 ± 0.026	2.475 ± 0.019	2.415 ± 0.020	0.337 ± 0.044
PI 95 σ	0.000 ± 0.000	0.100 ± 0.067	0.076 ± 0.070	0.119 ± 0.067	0.146 ± 0.023	0.144 ± 0.073	0.088 ± 0.002	0.076 ± 0.002	0.079 ± 0.002	0.066 ± 0.001	0.098 ± 0.011	0.058 ± 0.002	0.039 ± 0.002	0.053 ± 0.002	0.033 ± 0.003	0.119 ± 0.013

J7 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.453 ± 0.036	1.224 ± 0.171	1.321 ± 0.096	1.137 ± 0.010	0.255 ± 0.001	0.920 ± 0.002	0.874 ± 0.001	0.936 ± 0.001	0.885 ± 0.002	0.208 ± 0.004	0.944 ± 0.001	0.957 ± 0.054	0.961 ± 0.004	0.942 ± 0.035	0.114 ± 0.008
MAE	0.798 ± 0.025	0.743 ± 0.038	0.654 ± 0.159	0.654 ± 0.063	0.450 ± 0.035	0.416 ± 0.006	0.422 ± 0.007	0.398 ± 0.007	0.425 ± 0.007	0.398 ± 0.007	0.389 ± 0.010	0.413 ± 0.009	0.408 ± 0.015	0.418 ± 0.007	0.386 ± 0.027	0.411 ± 0.018
RMSE	0.903 ± 0.020	0.840 ± 0.046	0.654 ± 0.156	0.758 ± 0.064	0.557 ± 0.037	0.529 ± 0.011	0.528 ± 0.006	0.498 ± 0.007	0.532 ± 0.006	0.498 ± 0.006	0.498 ± 0.006	0.524 ± 0.015	0.517 ± 0.015	0.529 ± 0.016	0.496 ± 0.027	0.524 ± 0.016
NLL	1.541 ± 0.027	1.325 ± 0.044	1.064 ± 0.168	1.218 ± 0.099	0.961 ± 0.046	44.449 ± 5.922	0.910 ± 0.014	0.864 ± 0.023	0.916 ± 0.013	0.866 ± 0.020	75.968 ± 10.258	0.935 ± 0.035	0.899 ± 0.035	0.940 ± 0.035	0.884 ± 0.046	36.221 ± 12.337
QCE 95	0.014 ± 0.009	0.010 ± 0.006	0.008 ± 0.020	0.008 ± 0.010	0.010 ± 0.012	-0.768 ± 0.028	-0.015 ± 0.007	-0.007 ± 0.010	-0.012 ± 0.006	-0.002 ± 0.007	-0.791 ± 0.019	-0.016 ± 0.017	-0.012 ± 0.012	-0.014 ± 0.017	-0.008 ± 0.014	-0.749 ± 0.054
PI 95 μ	3.920 ± 0.000	3.467 ± 0.092	2.944 ± 0.327	3.275 ± 0.220	2.787 ± 0.090	0.321 ± 0.014	2.381 ± 0.008	2.297 ± 0.006	2.425 ± 0.004	2.324 ± 0.011	0.235 ± 0.022	2.432 ± 0.007	2.412 ± 0.026	2.475 ± 0.019	2.415 ± 0.020	0.337 ± 0.044
PI 95 σ	0.000 ± 0.000	0.056 ± 0.052	0.123 ± 0.075	0.109 ± 0.065	0.096 ± 0.026	0.168 ± 0.008	0.088 ± 0.002	0.076 ± 0.002	0.079 ± 0.002	0.066 ± 0.001	0.098 ± 0.011	0.058 ± 0.002	0.039 ± 0.002	0.053 ± 0.002	0.033 ± 0.003	0.119 ± 0.013

Poverty 26 03 18 07 02 - Full Training Set

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J6 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.487 ± 0.011	1.487 ± 0.006	1.468 ± 0.114	1.127 ± 0.045	0.290 ± 0.003	0.888 ± 0.003	0.874 ± 0.002	0.905 ± 0.001	0.892 ± 0.002	0.127 ± 0.007	0.881 ± 0.001	0.868 ± 0.002	0.896 ± 0.001	0.883 ± 0.002	0.116 ± 0.006
MAE	0.798 ± 0.025	0.798 ± 0.025	0.798 ± 0.025	0.722 ± 0.084	0.460 ± 0.026	0.423 ± 0.008	0.387 ± 0.008	0.382 ± 0.009	0.392 ± 0.007	0.385 ± 0.007	0.384 ± 0.008	0.387 ± 0.006	0.381 ± 0.007	0.392 ± 0.006	0.385 ± 0.008	0.385 ± 0.013
RMSE	0.903 ± 0.020	0.903 ± 0.020	0.903 ± 0.020	0.821 ± 0.087	0.568 ± 0.033	0.545 ± 0.011	0.499 ± 0.011	0.492 ± 0.011	0.503 ± 0.010	0.493 ± 0.010	0.490 ± 0.006	0.495 ± 0.012	0.487 ± 0.013	0.499 ± 0.012	0.490 ± 0.011	0.494 ± 0.012
NLL	1.541 ± 0.027	1.378 ± 0.022	1.373 ± 0.023	1.262 ± 0.081	0.985 ± 0.052	80.981 ± 18.642	0.881 ± 0.029	0.868 ± 0.030	0.883 ± 0.025	0.866 ± 0.027	16.635 ± 3.018	0.878 ± 0.029	0.865 ± 0.030	0.882 ± 0.027	0.867 ± 0.028	20.285 ± 3.723
QCE 95	0.014 ± 0.009	-0.004 ± 0.009	-0.004 ± 0.009	0.007 ± 0.020	0.017 ± 0.009	-0.798 ± 0.021	-0.016 ± 0.011	-0.014 ± 0.013	-0.012 ± 0.015	-0.008 ± 0.013	-0.681 ± 0.026	-0.012 ± 0.014	-0.010 ± 0.012	-0.008 ± 0.014	-0.005 ± 0.012	-0.704 ± 0.041
PI 95 μ	3.920 ± 0.000	3.563 ± 0.013	3.561 ± 0.008	3.398 ± 0.104	2.814 ± 0.164	0.244 ± 0.016	2.316 ± 0.008	2.290 ± 0.006	2.360 ± 0.005	2.340 ± 0.005	0.436 ± 0.043	2.301 ± 0.004	2.283 ± 0.008	2.340 ± 0.006	2.320 ± 0.009	0.392 ± 0.049
PI 95 σ	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.033 ± 0.031	0.121 ± 0.019	0.140 ± 0.009	0.044 ± 0.001	0.042 ± 0.001	0.038 ± 0.001	0.037 ± 0.001	0.133 ± 0.024	0.045 ± 0.002	0.043 ± 0.002	0.039 ± 0.002	0.036 ± 0.001	0.120 ± 0.022

J7 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.475 ± 0.000	1.475 ± 0.000	1.446 ± 0.067	1.476 ± 0.001	0.328 ± 0.001	0.888 ± 0.003	0.874 ± 0.002	0.905 ± 0.001	0.892 ± 0.002	0.127 ± 0.007	0.881 ± 0.001	0.868 ± 0.002	0.896 ± 0.001	0.883 ± 0.002	0.116 ± 0.006
MAE	0.798 ± 0.025	0.798 ± 0.025	0.798 ± 0.025	0.756 ± 0.078	0.798 ± 0.025	0.441 ± 0.013	0.387 ± 0.008	0.382 ± 0.009	0.392 ± 0.007	0.385 ± 0.007	0.384 ± 0.008	0.387 ± 0.006	0.381 ± 0.007	0.392 ± 0.006	0.385 ± 0.008	0.385 ± 0.013
RMSE	0.903 ± 0.020	0.903 ± 0.020	0.903 ± 0.020	0.861 ± 0.081	0.903 ± 0.020	0.563 ± 0.016	0.499 ± 0.011	0.492 ± 0.011	0.503 ± 0.010	0.493 ± 0.010	0.490 ± 0.006	0.495 ± 0.012	0.487 ± 0.013	0.499 ± 0.012	0.490 ± 0.011	0.494 ± 0.012
NLL	1.541 ± 0.027	1.380 ± 0.028	1.384 ± 0.033	1.318 ± 0.069	1.355 ± 0.020	282.055 ± 44.924	0.881 ± 0.029	0.868 ± 0.030	0.883 ± 0.025	0.866 ± 0.027	16.635 ± 3.018	0.878 ± 0.029	0.865 ± 0.030	0.882 ± 0.027	0.867 ± 0.028	20.285 ± 3.723
QCE 95	0.014 ± 0.009	-0.004 ± 0.009	-0.004 ± 0.009	0.002 ± 0.012	-0.004 ± 0.009	-0.866 ± 0.015	-0.016 ± 0.011	-0.014 ± 0.013	-0.012 ± 0.015	-0.008 ± 0.013	-0.681 ± 0.026	-0.012 ± 0.014	-0.010 ± 0.012	-0.008 ± 0.014	-0.005 ± 0.012	-0.704 ± 0.041
PI 95 μ	3.920 ± 0.000	3.577 ± 0.000	3.577 ± 0.000	3.514 ± 0.138	3.576 ± 0.001	0.152 ± 0.009	2.316 ± 0.008	2.290 ± 0.006	2.360 ± 0.005	2.340 ± 0.005	0.436 ± 0.043	2.301 ± 0.004	2.283 ± 0.008	2.340 ± 0.006	2.320 ± 0.009	0.392 ± 0.049
PI 95 σ	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.006 ± 0.012	0.000 ± 0.000	0.106 ± 0.010	0.044 ± 0.001	0.042 ± 0.001	0.038 ± 0.001	0.037 ± 0.001	0.133 ± 0.024	0.045 ± 0.002	0.043 ± 0.002	0.039 ± 0.002	0.036 ± 0.001	0.120 ± 0.022

Poverty 26 03 18 07 02 - Full Training Set

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J6 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.330 ± 0.109	1.326 ± 0.207	1.314 ± 0.171	1.113 ± 0.031	0.286 ± 0.008	0.888 ± 0.003	0.874 ± 0.002	0.905 ± 0.001	0.892 ± 0.002	0.127 ± 0.007	0.881 ± 0.001	0.868 ± 0.002	0.896 ± 0.001	0.883 ± 0.002	0.116 ± 0.006
MAE	0.798 ± 0.025	0.630 ± 0.054	0.678 ± 0.244	0.650 ± 0.095	0.472 ± 0.011	0.427 ± 0.006	0.387 ± 0.008	0.382 ± 0.009	0.392 ± 0.007	0.385 ± 0.007	0.384 ± 0.008	0.387 ± 0.006	0.381 ± 0.007	0.392 ± 0.006	0.385 ± 0.008	0.385 ± 0.013
RMSE	0.903 ± 0.020	0.745 ± 0.046	0.798 ± 0.267	0.753 ± 0.091	0.576 ± 0.015	0.533 ± 0.009	0.499 ± 0.011	0.492 ± 0.011	0.503 ± 0.010	0.493 ± 0.010	0.490 ± 0.006	0.495 ± 0.012	0.487 ± 0.013	0.499 ± 0.012	0.490 ± 0.011	0.494 ± 0.012
NLL	1.541 ± 0.027	1.216 ± 0.070	1.037 ± 0.064	1.188 ± 0.061	0.986 ± 0.025	231.332 ± 190.047	0.881 ± 0.029	0.868 ± 0.030	0.883 ± 0.025	0.866 ± 0.027	16.635 ± 3.018	0.878 ± 0.029	0.865 ± 0.030	0.882 ± 0.027	0.867 ± 0.028	20.285 ± 3.723
QCE 95	0.014 ± 0.009	-0.017 ± 0.033	-0.071 ± 0.170	0.004 ± 0.007	0.016 ± 0.010	-0.820 ± 0.052	-0.016 ± 0.011	-0.014 ± 0.013	-0.012 ± 0.015	-0.008 ± 0.013	-0.681 ± 0.026	-0.012 ± 0.014	-0.010 ± 0.012	-0.008 ± 0.014	-0.005 ± 0.012	-0.704 ± 0.041
PI 95 μ	3.920 ± 0.000	3.200 ± 0.268	3.087 ± 0.175	3.223 ± 0.219	2.815 ± 0.072	0.229 ± 0.098	2.316 ± 0.008	2.290 ± 0.006	2.360 ± 0.005	2.340 ± 0.005	0.436 ± 0.043	2.301 ± 0.004	2.283 ± 0.008	2.340 ± 0.006	2.320 ± 0.009	0.392 ± 0.049
PI 95 σ	0.000 ± 0.000	0.100 ± 0.067	0.076 ± 0.070	0.119 ± 0.067	0.146 ± 0.023	0.144 ± 0.073	0.044 ± 0.001	0.042 ± 0.001	0.038 ± 0.001	0.037 ± 0.001	0.133 ± 0.024	0.045 ± 0.002	0.043 ± 0.002	0.039 ± 0.002	0.036 ± 0.001	0.120 ± 0.022

J7 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.453 ± 0.036	1.224 ± 0.171	1.321 ± 0.096	1.137 ± 0.010	0.255 ± 0.001	0.888 ± 0.003	0.874 ± 0.002	0.905 ± 0.001	0.892 ± 0.002	0.127 ± 0.007	0.881 ± 0.001	0.868 ± 0.002	0.896 ± 0.001	0.883 ± 0.002	0.116 ± 0.006
MAE	0.798 ± 0.025	0.743 ± 0.038	0.554 ± 0.159	0.654 ± 0.063	0.450 ± 0.035	0.416 ± 0.006	0.387 ± 0.008	0.382 ± 0.009	0.392 ± 0.007	0.385 ± 0.007	0.384 ± 0.008	0.387 ± 0.006	0.381 ± 0.007	0.392 ± 0.006	0.385 ± 0.008	0.385 ± 0.013
RMSE	0.903 ± 0.020	0.840 ± 0.046	0.654 ± 0.156	0.758 ± 0.064	0.557 ± 0.037	0.529 ± 0.011	0.499 ± 0.011	0.492 ± 0.011	0.503 ± 0.010	0.493 ± 0.010	0.490 ± 0.006	0.495 ± 0.012	0.487 ± 0.013	0.499 ± 0.012	0.490 ± 0.011	0.494 ± 0.012
NLL	1.541 ± 0.027	1.325 ± 0.044	1.064 ± 0.168	1.218 ± 0.099	0.961 ± 0.046	44.449 ± 5.922	0.881 ± 0.029	0.868 ± 0.030	0.883 ± 0.025	0.866 ± 0.027	16.635 ± 3.018	0.878 ± 0.029	0.865 ± 0.030	0.882 ± 0.027	0.867 ± 0.028	20.285 ± 3.723
QCE 95	0.014 ± 0.009	0.010 ± 0.006	0.008 ± 0.020	0.008 ± 0.010	0.010 ± 0.012	-0.768 ± 0.028	-0.016 ± 0.011	-0.014 ± 0.013	-0.012 ± 0.015	-0.008 ± 0.013	-0.681 ± 0.026	-0.012 ± 0.014	-0.010 ± 0.012	-0.008 ± 0.014	-0.005 ± 0.012	-0.704 ± 0.041
PI 95 μ	3.920 ± 0.000	3.467 ± 0.092	2.944 ± 0.327	3.275 ± 0.220	2.787 ± 0.060	0.321 ± 0.014	2.316 ± 0.008	2.290 ± 0.006	2.360 ± 0.005	2.340 ± 0.005	0.436 ± 0.043	2.301 ± 0.004	2.283 ± 0.008	2.340 ± 0.006	2.320 ± 0.009	0.392 ± 0.049
PI 95 σ	0.000 ± 0.000	0.056 ± 0.052	0.123 ± 0.075	0.109 ± 0.065	0.096 ± 0.026	0.168 ± 0.006	0.044 ± 0.001	0.042 ± 0.001	0.038 ± 0.001	0.037 ± 0.001	0.133 ± 0.024	0.045 ± 0.002	0.043 ± 0.002	0.039 ± 0.002	0.036 ± 0.001	0.120 ± 0.022